

ORF 524 - Fall 2026

Week 03

$$MSE = E_{\theta} [(T(x) - \beta(\theta))^2] \quad X \sim P_{\theta}, \quad \beta(\theta) = \beta(P_{\theta})$$

$$= E_{\theta} [(T(x) - E_{\theta}[T(x)] + E_{\theta}[T(x)] - \beta(\theta))^2]$$

$$= \underbrace{E_{\theta} [(T(x) - E_{\theta}[T(x)])^2]}_{V_{\theta}[T(x)]} + 0 + \underbrace{(E_{\theta}[T(x)] - \beta(\theta))^2}_{\text{bias}}$$

$$= V_{\theta}[T(x)] + \text{Bias}_{\theta}^2(T(x), \beta(\theta))$$

- Unconstrained

$$T^*(\cdot) \in \underset{T \in \mathcal{T}}{\operatorname{argmin}} \mathbb{E}_{\theta} \left[(T(x) - \beta(\theta))^2 \right]$$

- Constrained

$$T^{\Delta}(\cdot) \in \underset{T \in \mathcal{T}_0}{\operatorname{argmin}} \mathbb{E}_{\theta} \left[(T(x) - \beta(\theta))^2 \right]$$

$$\mathcal{T}_0 = \left\{ T(\cdot) : \mathbb{E}_{\theta} [|T(x)|^2] < \infty, \right.$$

$$\left. \mathbb{E}_{\theta} [T(x)] = \beta(\theta), \forall \theta \in \Theta \right\}$$

$$X \sim P_\theta \quad P_\theta \in \mathcal{P}$$

- $\hat{\theta}(x)$ is an unbiased estimator $\Leftrightarrow$
 - $\hat{\theta}(x)$ not function of θ
- $\tilde{\theta}(x) = \mathbb{E}_\theta[\hat{\theta}(x) | T(x)]$
 - $\mathbb{E}_\theta[\hat{\theta}(x)] = \theta$
- $T(x)$ is sufficient statistic $\Leftrightarrow P_{X|T(x); \theta} = P_{X|T(x)}$

① $\tilde{\theta}(x)$ is (also) an unbiased estimator.

$$\textcircled{2} \quad V_\theta[\tilde{\theta}(x)] \leq V_\theta[\hat{\theta}(x)]$$

Proof of ① : (i) $\tilde{\theta}(x) = \mathbb{E}_\theta[\hat{\theta}(x) | T(x)]$

(ii) $\mathbb{E}_\theta[\tilde{\theta}(x)] = \mathbb{E}_\theta[\mathbb{E}[\hat{\theta}(x) | T(x)]] = \mathbb{E}_\theta[\hat{\theta}(x)] = \theta$

Proof of ②: Recall $V[A] = \mathbb{E}[V[A|B]] + V[\mathbb{E}[A|B]]$

$$A = \hat{\Theta}(x)$$

$$B = T(x)$$

$$V_{\theta}[\hat{\Theta}(x)] = \mathbb{E}_{\theta}[V[\hat{\Theta}(x) | T(x)]] + V_{\theta}[\underbrace{\mathbb{E}[\hat{\Theta}(x) | T(x)]}_{\hat{\tilde{\Theta}}(x)}]$$

$$= \textcircled{+} + V_{\theta}[\hat{\tilde{\Theta}}(x)]$$

likelihood theory

$$X_1, X_2, \dots, X_n \sim P_{\theta_0}$$

$$P_{\theta_0} \in \mathcal{P} = \{ P_{\theta} : \theta \in \Theta \subset \mathbb{R}^d \}$$

- Density: $f_{\underline{x}}(\underline{x}; \theta)$ + iid.

$$f_{\underline{x}}(\underline{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta) \Rightarrow \text{loglik} = l_n(\theta)$$

$$l_n(\theta) = \log \left(\prod_{i=1}^n f_{X_i}(x_i; \theta) \right) = \sum_{i=1}^n \log f_{X_i}(x_i; \theta)$$

- More generally, $l_n(\theta) = \log f_{\underline{x}}(\underline{x}; \theta)$

- Score: $\frac{\partial}{\partial \theta} l_n(\theta) = \frac{\partial}{\partial \theta} \log f_{\underline{x}}(\underline{x}; \theta) \stackrel{\text{iid}}{\downarrow} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f_{X_i}(x_i; \theta)$

Result 1 : $\mathbb{E}_{\theta} \left[\frac{\partial}{\partial \theta} \log f(x; \theta) \right] = 0 \quad \forall \theta \in \Theta$

$$1 = \int f(x; \theta) d\mu(x) \quad \forall \theta \in \Theta$$

$$\frac{\partial}{\partial \theta} 1 = 0 = \frac{\partial}{\partial \theta} \int f(x; \theta) d\mu(x)$$

crazy
line

$$= \int \left[\frac{\frac{\partial}{\partial \theta} f(x; \theta)}{f(x; \theta)} \right] \cdot f(x; \theta) \cdot d\mu(x)$$

$$= \int \left[\frac{\partial}{\partial \theta} \log f(x; \theta) \right] f(x; \theta) d\mu(x)$$

Note :

$$\theta_0 = \underset{\theta \in \Theta}{\operatorname{argmin}} \mathbb{E}_{\theta_0} [\log f_X(x; \theta)]$$

$$\Rightarrow \mathbb{E}_{\theta_0} [S_{\theta}(x)] = 0 \quad \text{if (and only if)} \\ \theta = \theta_0$$

$$S_{\theta}(x) = \frac{\partial}{\partial \theta} \log f_X(x; \theta)$$

Result 2

$$0 = \mathbb{E}_{\theta} \left[\frac{\partial}{\partial \theta} \log f(x; \theta) \right] \quad \forall \theta \in \Theta$$

$$\frac{\partial}{\partial \theta^T} 0 = 0 = \frac{\partial}{\partial \theta^T} \mathbb{E}_{\theta} \left[\frac{\partial}{\partial \theta} \log f(x; \theta) \right]$$

be careful

$$= \frac{\partial}{\partial \theta^T} \int \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right) f(x; \theta) d\mu(x)$$


$$= \int \left(\frac{\partial}{\partial \theta \partial \theta^T} \log f(x; \theta) \right) f(x; \theta) d\mu(x)$$

$$+ \int \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right) \frac{\partial}{\partial \theta^T} f(x; \theta) d\mu(x)$$

thus :

$$- E_{\theta} \left[\frac{\partial}{\partial \theta} \log f(x|\theta) \right] = E_{\theta} \left[S_{\theta}(x) S_{\theta}(x)^T \right]$$

$$= V_{\theta} [S_{\theta}(x)]$$

$$= I(\theta)$$

CRLB

$$\textcircled{1} \quad |\mathbb{E}[QY]| \leq \mathbb{E}[|QY|] \leq \left(\mathbb{E}[|Q|^2]\right)^{1/2} \left(\mathbb{E}[|Y|^2]\right)^{1/2}$$

$$\textcircled{2} \quad Q = W - \mathbb{E}[W] \quad Y = Z - \mathbb{E}[Z]$$

$$|\text{Cov}(W, Z)| \leq \left(\text{Var}[W]\right)^{1/2} \left(\text{Var}[Z]\right)^{1/2}$$

$$\textcircled{3} \quad W = S_\theta(x) \Rightarrow \mathbb{E}_\theta[S(x)] = 0 \Rightarrow Q = W$$

$$Z = T(x), \quad T(x) \text{ is some statistic}$$

$$V_{\theta}[z] = V_{\theta}[T(x)]$$

$$\geq \frac{\text{Cov}_{\theta}(S_{\theta}(x), T(x))^2}{V_{\theta}[S_{\theta}(x)]} = \frac{\left(\frac{\partial}{\partial \theta} \mathbb{E}_{\theta}[T(x)]\right)^2}{\mathcal{I}(\theta)}$$

Note :

$$\text{Cov}_{\theta}(S_{\theta}(x), T(x)) = \mathbb{E}_{\theta}[S_{\theta}(x) T(x)]$$

$$= \int \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right) \cdot T(x) \cdot f(x; \theta) d\mu(x)$$

$$= \int \frac{\frac{\partial}{\partial \theta} f(x; \theta)}{\cancel{f(x; \theta)}} T(x) \cancel{f(x; \theta)} d\mu(x)$$

$$= \frac{\partial}{\partial \theta} \int T(x) f(x; \theta) d\mu(x) = \frac{\partial}{\partial \theta} \mathbb{E}_{\theta}[T(x)]$$